

VAANISETU

Technical Analysis & Quality Assurance Report

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Prepared for: BAIF Development Research Foundation (Pune HQ)

Deployment Environment: 100% Offline (Zero-Connectivity)

CONFIDENTIALITY NOTE: This document contains automated analysis of the VaaniSetu pipeline, model benchmarks, and QA edge-case testing results. Generated locally on the deployment machine.

1. AI Model Selection & Benchmarking

To ensure VaaniSetu meets BAIF's strict requirements for rural deployment, we conducted a rigorous comparative analysis of available AI translation models. The primary constraints were: 100% offline capability, deep support for official Indian languages (specifically tribal/rural dialects), and acceptable computational overhead.

Model	Offline Ready?	Indian Languages	Performance (BLEU Estimate)
Google Cloud API	No (Fails Constraint)	11 Languages	High
Meta NLLB-200	Yes (~2.4 GB)	Global focus	Moderate on Indian dialects
IndicTrans2 (Chosen)	Yes (~1.5 GB)	22 Languages	SOTA (~37 BLEU for Hindi/Marathi)

Decision Justification:

IndicTrans2 was definitively selected because it was explicitly trained by AI4Bharat (IIT Madras) on Indian Government corpora. It drastically outperforms Meta's NLLB on low-resource languages relevant to BAIF (such as Bodo, Santhali, and Maithili).

2. Quality Assurance & Edge-Case Testing

Hackathon prototypes often fail in production because they only test the 'Happy Path'. VaaniSetu was subjected to rigorous stress testing simulating actual NGO field constraints.

QA-01 (Happy)	Full Video Pipeline Translation (English -> Marathi)	PASSED
QA-02 (Happy)	Reverse Bridge Farmer Audio (Marathi -> English)	PASSED
EDGE-01	Memory Spike: 2GB upload via chunked 64KB stream	PASSED
EDGE-02	Disk Exhaustion: Workspace & Original file auto-purge	PASSED
EDGE-03	AI Hallucination: Whisper pure-silence sanitization	PASSED
EDGE-04	Cache Collision: Language-aware Smart Deduplication	PASSED

Key Defensive Engineering Results:

- The 64KB chunk-streaming implementation ensures RAM usage stays flat during massive file uploads, protecting older laptops from crashing.
- The Auto-Disk Recovery function successfully purged 100% of intermediate gigabyte-heavy uncompressed .wav files post-job.
- The Confidence Gate successfully flagged translations scoring below 0.65 threshold to the manual Review Queue.

3. Last-Mile Output Format Validation

VaaniSetu is not a 'web app'. It is an offline engine designed to generate physical and distributed outputs for the last mile. The following outputs have been generated and validated:

Format	Target NGO Distribution Channel
Bilingual .docx	Physical printed handouts by field officers
IVR Audio (.wav)	8kHz mono output for direct feature phone telecom broadcast
WhatsApp (.mp4)	Auto-chunked <15MB to bypass strict WhatsApp sharing limits
Captioned Video	Gram Panchayat community TV / projector screenings

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